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A conceptualization and empirical investigation of strategic sourcing approaches for engineer-to-order manufacturers

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ABSTRACT

Since supplier capabilities often are essential for the success of a new product, previous studies suggest manufacturers to integrate strategic sourcing into the early stages of their new product development (NPD) processes. However, the literature on this topic offers only limited guidance for manufacturers in the engineer-to-order (ETO) context, where late strategic sourcing can be inevitable due to, for example, uncertainty or changing customer requirements. To address this gap in the literature, this article first draws on the literature to conceptualize five distinct strategic sourcing approaches that ETO manufacturers can employ. The approaches differ in terms of the timing of three strategic sourcing activities during the NPD process: make-or-buy analysis, supplier selection and supplier collaboration. To investigate the approaches, we conducted an embedded single case study involving a large turbine manufacturer. The study supports the existence of all the strategic sourcing approaches and identifies distinct conditions associated with each of them. Thereby, this article extends the literature with additional clarity about strategic sourcing in the ETO context, as well as when different approaches occur.

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1. Introduction

Engineer-to-order (ETO) products are characterized by being high-cost, engineering intensive, ordered in low volumes, and developed according to customer requirements (Hobday 2000; Willner et al. 2016; Acha et al. 2004; Gosling and Naim 2009). Due to these characteristics, the development of new ETO products often requires the involvement of suppliers and the sourcing of many items and services (Willner et al. 2016; Cannas et al. 2022). To fully capitalize on supplier capabilities and reduce the complexity stemming from supply, the literature suggests manufacturers to integrate the purchasing function early into new product development (NPD) processes (Reitsma, et al. 2023; Mikkelsen and Johnsen 2019; Schiele 2010). Together with the research and development (R&D) function, the purchasing function can engage in strategic sourcing, which aims at facilitating the involvement of suppliers with desirable capabilities and preventing issues such as short-termism and outsourcing critical items to cheap but unreliable and globally dispersed suppliers (Shishodia, Verma, and Jain 2022; Li et al. 2023; Reitsma, Hilletofth, Johansson 2023).

Despite the many valuable insights provided by previous NPD sourcing literature, there still appears to be a need to investigate how ETO manufacturers engage in strategic sourcing during NPD. Specifically, the ETO context involves high levels of product complexity, uncertainty and customization (Gosling and Naim 2009; Hobday, Rush, and Tidd 2000; Willner et al. 2016; Cannas and Gosling 2021), which may lead to sourcing challenges that are different from those in, for example, make-to-stock or assemble-to-order contexts (Seth and Rastogi 2018; Gosling, Hewlett, and Naim 2020a, 2020b). Furthermore, due to the engineering-intensive nature of ETO products, their NPD processes can overlap, be complex (Bhalla, Alfnes, and Hvolby 2023; Iakymenko et al. 2022), and sometimes span more than one decade (Reitsma et al. 2023). Moreover, during these processes, the customer-specific requirements can be difficult to define or change over time (Willner et al. 2016; Hobday, Rush, and Tidd 2000). This forces the R&D and purchasing functions of ETO manufacturers to continuously adapt their work, potentially causing a shift from early to late strategic sourcing during NPD (Emblemsvåg 2014; Hicks, McGovern, and Earl 2000; Vaagen, Kaut, and Wallace 2017). However, while the ETO literature (e.g. Shurrab, Jonsson, and Johansson 2022) suggests that strategic sourcing may occur on different occasions, there has been no examination of the conditions affecting its timing during NPD.

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The discussion above suggests a lack of understanding of the conditions that drive early or late strategic sourcing in the ETO context. This gap gives rise to the following research aims:

1. To conceptualize distinct strategic sourcing approaches in terms of the timing of sourcing activities during an ETO manufacturer's NPD process.
2. To investigate the existence of the identified strategic sourcing approaches.
3. To identify conditions associated with each of the strategic sourcing approaches.

To achieve these aims, the present article first uses the literature to define the strategic sourcing process – including timing alternatives for its key activities during NPD – based on which five strategic sourcing approaches that may be applicable for ETO manufacturers are conceptualized. Afterwards, to investigate the existence and conditions of the five approaches, a case study involving an ETO manufacturer of large turbines was conducted. The study thereby contributes to the literature by identifying the conditions that warrant early or late sourcing in the ETO context, and to practice by providing ETO manufacturers with insights into different timing alternatives for strategic sourcing activities.

The remainder of the article is structured as follows. To begin with, a literature review of strategic sourcing during NPD is presented in Section 2, after which five strategic sourcing approaches for ETO manufacturers are conceptualized in Section 3. This is followed by a description of the research design used to empirically explore the five approaches in Section 4. The empirical data are analysed in Section 5. Finally, Section 6 discusses the empirical findings in the context of previous literature, and Section 7 presents conclusions, limitations and future research avenues.

2. Literature review

To provide a basis for achieving the research aims, this section first discusses the literature on strategic sourcing during NPD. Hereafter, the section discusses three strategic sourcing 'activities' that can be performed at different points in time during the NPD process.

2.1. Strategic sourcing during NPD

Although the literature does not provide fully consistent results, supplier involvement in NPD processes has been recognized as a potential source of competitive advantage (Luzzini et al. 2015; Suurmond, Wynstra, and Dul 2020). Suppliers can be invited to join NPD projects and may have a key role in improving manufacturers' innovation performance (Clark 1989; Handfield et al. 1999; Bonaccorsi and Lipparini 1994). The ETO literature also highlights the potential benefits of supplier involvement in NPD. For example, Mello, Strandhagen, and Alfnes (2015) argue that supplier involvement allows tackling various problem situations in meeting customer-specific requirements. However, supplier involvement can be a difficult task since suppliers with the necessary experience and capabilities may be unavailable or not interested in contributing to NPD projects (Monczka 2000; Schiele 2006).

Since the purchasing (or procurement) function has become the common interface with suppliers, it can play a key role in facilitating supplier involvement in NPD (Araujo, Dubois, and Gadde 1999; Ellram and Pearson 1993). For example, the purchasing function can contribute to the NPD team with its knowledge of supplier capabilities and its relationships with suppliers in communicating the requirements of the R&D function for a new product externally (Mikkelsen and Johnsen 2019). When joining the NPD team, the purchasing function may require 'purchasing engineers' (Burt and Soukup 1985) with a strong technical background and the ability to work with the R&D function from an early stage (Schiele 2010).

In the ETO context, the NPD team consisting of staff from the R&D and purchasing functions can facilitate the involvement of suppliers with desirable capabilities by collaborating on three strategic sourcing activities: (1) make-or-buy analysis, (2) supplier selection and (3) supplier collaboration (Reitsma et al. 2023). First, a make-or-buy analysis focuses on identifying the strategic nature of a product item or service and the level of specialization in determining the outcome of the analysis (Venkatesan 1992; Kistruck et al. 2015). Second, supplier selection entails selecting sources of supply in a manner that makes them an integral part of the buying organization for the product items or services they are to provide (Zenz and Thompson 1994). Third, supplier collaboration has been described as collaborating with suppliers, allowing buying organizations to obtain complementary capabilities without capital investments (Carter and Narasimhan 1990). Research that specifically focuses on NPD explores how these sourcing activities are made for new products (Wouters et al. 2009) and has shown that they can have a substantial impact on NPD performance (Claypool, Norman, and Needy 2014; Monczka and Trent 1991).

Based on the three perspectives described above, this study conceptualizes strategic sourcing as a process involving the activities of make-or-buy analysis, supplier selection and supplier collaboration, and focuses on their timing in the NPD process. Given the lack of ETO-oriented literature on the topic, the remainder of this section draws on the literature to elaborate on the strategic sourcing activities, as well as their potential timing in relation to different time periods (i.e. stages) of the NPD process.

2.2. Make-or-buy analysis

Make-or-buy analysis is a key strategic activity of NPD teams (Ellram and Carr 1994; Holcomb and Hitt 2007), which focuses on balancing internal and external sourcing (Fine, Golany, and Naseraldin 2005; Fine 2000). This concerns both the physical items (e.g. components, modules) required for a new product and services associated with items (e.g.

research and technology development, manufacturing operations) (Fine, Golany, and Naseraldin 2005; Wynstra, Van Weele, and Axelsson 1999; Wynstra, Weggeman, and van Weele 2003). During a make-or-buy analysis, consideration should be given to various factors, including the competitive importance and lead time or cost of an item or service, as well as the risk of supplier dependency (Noori and Georgescu 2008; Wynstra, Van Weele, and Axelsson 1999). As an example, 'buying' is appropriate when specialist suppliers have economies of scale in a range of business areas and superior capabilities that are difficult or costly to replicate (McIvor 2009). Alternatively, 'making' is appropriate when there are few suppliers and an organization has a strong performance position in relation to certain items or services, or when they are critical for achieving a competitive advantage and involve intellectual property rights (Noori and Georgescu 2008; McIvor 2009). The identification of which item or service to keep in-house involves assessing the availability of internal capabilities and those of candidate suppliers (Fine 2000; Noori and Georgescu 2008; Wynstra, Van Weele, and Axelsson 1999; Wynstra, Weggeman, and van Weele 2003). For example, a make-or-buy analysis can consider whether overall NPD costs, risks or lead times can be reduced by internally sourcing items or services (Fine 2000; Nepal, Monplaisir, and Famuyiwa 2011), as well as the capital investments required to bridge the potential performance disparity between the manufacturer and its candidate suppliers (McIvor 2009).

The literature provides alternative perspectives on when to complete the make-or-buy analysis in NPD processes (Table 1). One example is Wynstra, Van Weele, and Axelsson (1999), who recommend completing the analysis before the concept design stage of NPD, as this saves time and effort in subsequent stages. Others suggest completing the analysis when creating and evaluating new product ideas, which occurs during the concept design stage of NPD (Chiu and Kremer 2014; Noori and Georgescu 2008; Chiu and Okudan 2014). Finally, some researchers suggest completing the make-or-buy analysis during the detail design stage of NPD, when the product bill-of-material (BOM) becomes available (Fine, Golany, and Naseraldin 2005; Nepal, Monplaisir, and Famuyiwa 2011). For example, this allows performing make-or-buy analyses based on choices made concerning design geometrics and material selection (Shurrab, Jonsson, and Johansson 2022).

2.3. Supplier selection

Supplier selection concerns the actual choice of the suppliers of the items or services needed for a new product (Fine 2000; Fine, Golany, and Naseraldin 2005; Hsu et al. 2006). Such choices typically require evaluations of whether supplier capabilities align with the characteristics of the outsourced items or services (Petersen, Handfield, and Ragatz 2005; Song and Di Benedetto 2008). As an example, when an item is critical for achieving a competitive advantage, suppliers should be innovative and technologically capable (Brewer and Arnette 2017; Di Benedetto et al. 2003; Noori and Georgescu 2008; Gosling, Hewlett, and Naim 2020a). Other common supplier-selection criteria include cost, quality, flexibility and delivery performance (e.g. Chiu and Kremer 2014; Claypool, Norman, and Needy 2014; Fine, Golany, and Naseraldin 2005; Sabri, Micheli, and Cagno 2022). During supplier selection, the potential benefit of selecting multiple suppliers for a specific item or service should also be considered (Wynstra, Weggeman, and van Weele 2003). This involves comparing the costs and benefits obtained from the use of economies of scale associated with single sourcing with the costs and benefits of multiple sourcing (Claypool, Norman, and Needy 2014; Noori and Georgescu 2008; Dowlatshahi 1996; Krikke, Bloemhof-Ruwaard, and Van Wassenhove 2003). When a manufacturer also intends to collaborate closely with suppliers during early product design activities, it is crucial to select suppliers based on their inclination for openness in a relationship, their track record with other customers, and their knowledge and understanding of the manufacturer (Gallear et al. 2022). Furthermore, while manufacturers can benefit from collaborating with both familiar and unfamiliar suppliers, the selection of familiar suppliers may be preferred because of the ease at which they can be involved in NPD (Subramanian and Soh 2017).

The literature includes different suggestions for when to select suppliers in the NPD process (Table 1). First, Wynstra, Van Weele, and Axelsson (1999) propose selecting suppliers before the concept design stage of NPD. This enables early identification of key suppliers that can contribute to NPD

Table 1. The NPD stages at which strategic sourcing activities can occur.

Strategic sourcing activity	NPD stage	References
Make-or-buy analysis	Pre-concept design	Wynstra, Van Weele, and Axelsson (1999)
	Concept design	Chiu and Kremer (2014); Chiu and Okudan (2014); Noori and Georgescu (2008)
	Detail design	Fine, Golany, and Naseraldin (2005); Nepal, Monplaisir, and Famuyiwa (2011); Shurrab, Jonsson, and Johansson (2022)
Supplier selection	Pre-concept design	Wynstra, Van Weele, and Axelsson (1999)
	Concept design	Chiu and Kremer (2014); Noori and Georgescu (2008)
	Detail design	Claypool, Norman, and Needy (2014); Fine, Golany, and Naseraldin (2005); Gokhan, Needy, and Norman (2010); Graves and Willems (2005); Nepal, Monplaisir, and Famuyiwa (2011); Pham and Yenradee (2017)
Supplier collaboration	Concept design	Chiu and Kremer (2014); Clark (1989); Koufteros, Edwin Cheng, and Lai (2007); Noori and Georgescu (2008); Petersen, Handfield, and Ragatz (2005); Primo and Amundson (2002); Ragatz, Handfield, and Petersen (2002); Ragatz, Handfield, and Scannell (1997)
	Detail design	Fine, Golany, and Naseraldin (2005); Noori and Georgescu (2008); Monczka and Trent (1991)

with, for example, new ideas or technologies (Wynstra, Van Weele, and Axelsson 1999). Another recommended timing of supplier selection is during the concept design stage of NPD (Chiu and Kremer 2014; Noori and Georgescu 2008). This introduces higher flexibility and more time to prepare and respond to potential supply challenges (Chiu and Kremer 2014). A final proposed timing of supplier selection is during the detail design stage of NPD (Fine, Golany, and Naseraldin 2005; Graves and Willems 2005; Pham and Yenradee 2017; Nepal, Monplaisir, and Famuyiwa 2011). This allows the creation of an optimal match between the characteristics of suppliers and the structure of the product's design(s) listed in a BOM (Fine, Golany, and Naseraldin 2005; Graves and Willems 2005). For example, the BOM can be used to choose the best design alternative to manufacture for each item and which supplier to select based on the impact of a potential supplier on lead times and costs (Gokhan, Needy, and Norman 2010; Claypool, Norman, and Needy 2014).

2.4. Supplier collaboration

Supplier collaboration refers to the involvement of suppliers in product design activities (Fine 2000; Fine, Golany, and Naseraldin 2005). The closeness of a collaboration depends on the division of labour and tasks between the manufacturer and its supplier as measured by supplier design responsibility (Azadegan and Dooley 2010; Clark 1989; Wynstra et al. 2012). When products are critical or highly innovative, a manufacturer may need to rely heavily on suppliers with superior design capabilities (Picaud-Bello et al. 2019; Song and Thieme 2009; Petersen, Handfield, and Ragatz 2005; Fine, Golany, and Naseraldin 2005; Noori and Georgescu 2008; Ülkü and Schmidt 2011). By gaining access to these superior, external capabilities and delegating design responsibilities to suppliers, a manufacturer requires less development and engineering resources (Clark 1989). This may also allow quicker responses to changing customer demands, which, according to previous studies, ultimately reduces NPD costs (Clark 1989; Ragatz, Handfield, and Scannell 1997; Handfield et al. 1999). However, it should be noted that Suurmond, Wynstra, and Dul (2020) found that the literature does not agree that the delegation of design responsibilities to suppliers yields higher product quality, financial performance or product innovativeness. Furthermore, with close supplier collaboration, the main challenge is the buyer and supplier can become mutually dependent, placing themselves at risk of opportunistic behaviour by their partners (Badaracco 2009; Kaufman, Wood, and Theyel 2000). However, close supplier collaboration is not always needed during NPD. For example, relatively superficial relationships may suffice when working with suppliers that are responsible for non-critical items or services (Picaud-Bello et al. 2019; Gosling, Hewlett, and Naim 2020b; Petersen, Handfield, and Ragatz 2005).

Supplier collaboration can start at various times during the NPD process (Table 1). In this context, it should be noted that the moment at which suppliers become involved in NPD is not necessarily the same moment that supplier selection is completed since manufacturers may want to 'test' collaborative relationships before making financial transactions (Cole and Aitken 2019). Collaboration with key suppliers can start already during the concept design stage of NPD (Chiu and Kremer 2014; Noori and Georgescu 2008). This allows the buyer to acquire more ideas and concepts from suppliers, potentially leading to product innovation benefits (Koufteros, Edwin Cheng, and Lai 2007), increased manufacturability (Clark 1989), shorter time-to-market (Ragatz, Handfield, and Petersen 2002), improved product quality (Ragatz, Handfield, and Scannell 1997; Primo and Amundson 2002), reduced NPD cost (Primo and Amundson 2002) and higher process alignment as well as a privileged relationship (Petersen, Handfield, and Ragatz 2005). However, other studies argue that early supplier collaboration can affect NPD negatively. For example, discussing new product ideas and concepts with suppliers requires effective information and knowledge sharing, which is costly to manage, and may slow down the overall NPD process (Hartley, Zirger, and Kamath 1997; Wynstra et al. 2012). Furthermore, suppliers do not necessarily have the capability or capacity to contribute to the early stages of NPD (Moradlou, Roscoe, and Ghadge 2022). Therefore, another proposed timing of supplier collaboration is during the detail design stage of NPD (Fine, Golany, and Naseraldin 2005; Noori and Georgescu 2008; Monczka and Trent 1991). At this point in time, a more detailed product BOM enables forming ties with suppliers based on known product characteristics, including, for example, the level of modularity or innovativeness (Fine, Golany, and Naseraldin 2005; Noori and Georgescu 2008).

3. Towards a typology of strategic sourcing in the ETO context

As shown in the preceding sections, the literature describes various approaches to strategic sourcing during NPD. However, as also argued in these sections, the ETO context tends to be ignored in the literature (Gosling and Naim 2009; Hobday, Rush, and Tidd 2000; Cannas and Gosling 2021). In the ETO context, NPD processes tend to be complex and lengthy (Reitsma et al. 2023) due to involving customer-specific requirements that are difficult to define or change over time (Willner et al. 2016; Hobday, Rush, and Tidd 2000). This forces ETO manufacturers to continuously adjust their strategic sourcing activities (Emblemsvåg 2014; Hicks, McGovern, and Earl 2000; Vaagen, Kaut, and Wallace 2017). Thus, while the literature (e.g. Clark 1989; Koufteros, Rawski, and Rupak 2010; Ragatz, Handfield, and Petersen 2002) often investigates supplier involvement in relation to the early NPD stages, the characteristics of ETO products may make late sourcing inevitable for manufacturers. For example, changing customer requirements or emerging product properties may force ETO manufacturers to suddenly outsource and involve suppliers in the late stages of NPD (Willner et al. 2016; Hobday, Rush, and Tidd 2000). To address the sparseness of knowledge on the conditions that drive early or late strategic sourcing in the ETO context, the present section adds detail to previous studies by developing

a typology for ETO manufacturers. Guided by the literature discussed in Section 2, essential components of the typology are the following strategic sourcing activities: (1) make-or-buy analysis, (2) supplier selection and (3) supplier collaboration.

As shown in Section 2 and Table 1, the three strategic sourcing activities can occur in different NPD stages: pre-concept design, concept design, or detail design. While the pre-concept and concept design stages can be referred to as 'early' timing, the detail design stage can be referred to as 'late' timing (Chiu and Kremer 2014). By distinguishing between early and late in the three strategic sourcing activities, eight combinations can be made, of which only five are realistic. Specifically, the three variants involving 'late make-or-buy analysis' followed by early timing in one or both the subsequent activities are not feasible, since the make-or-buy analysis is typically regarded as the start of the strategic sourcing process (Claypool, Norman, and Needy 2014; Fine, Golany, and Naseraldin 2005; Graves and Willems 2005). Hence, the proposed typology assumes that if the make-or-buy analysis is performed late during NPD, so too are the other activities.

Based on the above discussion, Table 2 presents a typology including five distinct strategic sourcing approaches in the ETO context. These approaches are subsequently discussed.

First, the EEE approach entails performing all strategic sourcing activities early in the NPD process. Despite the ETO context typically involving high levels of uncertainty (Willner et al. 2016), there may be situations where characteristics of items or services are known or easy to anticipate. In such situations, the EEE approach may occur, especially when early supplier collaboration is essential to managing high levels of product complexity or meeting customer requirements, which is often the case in the ETO context (Cannas and Gosling 2021; Willner et al. 2016; Hobday, Rush, and Tidd 2000). Second, the ELE approach differs from that of EEE by delaying supplier selection until late in the NPD process. Since the ETO context involves unpredictable and emergent product properties (Willner et al. 2016; Hobday, Rush, and Tidd 2000), late supplier selection can be associated with benefits such as maintaining product design flexibility and multiple sourcing. However, early supplier collaboration may still be desired when early input and supplier expertise helps in the identification of potential product design improvements. Third, the EEL approach differs from that of EEE by delaying supplier collaboration until late during NPD. This approach may occur when the late NPD stages are dependent on the input of a few critical suppliers that are difficult to replace. In such situations, it may be preferable to select suppliers as early as possible, although supplier collaboration is not needed until late in the NPD process. Fourth, the ELL approach entails performing the make-or-buy analysis at an early stage of NPD, while delaying supplier selection and collaboration until late in the NPD process. Since NPD in the ETO context typically takes many years to complete and involves a large number of items (Willner et al. 2016; Hobday, Rush, and Tidd 2000), an early make-or-buy analysis may help in anticipating which suppliers to select and collaborate with during the later stages of NPD. Finally, the LLL approach entails performing all sourcing activities late in the NPD process. This approach may emerge when precise characteristics of items or services are unknown until late in the NPD process. Such situations can be the result of changing customer requirements, emerging product properties, or unexpected events (Willner et al. 2016; Hobday, Rush, and Tidd 2000).

Table 2. Typology of strategic sourcing for ETO manufacturers.

Strategic sourcing approach				
#	Make-or-buy analysis	Supplier selection	Supplier collaboration	Abbreviation
1	Early	Early	Early	EEE
2	Early	Late	Early	ELE
3	Early	Early	Late	EEL
4	Early	Late	Late	ELL
5	Late	Late	Late	LLL

Note: Early = During the pre-concept design or concept design stage of NPD; Late = During the detail design stage of NPD.

4. Research design

4.1. Case study methodology

To investigate the existence of the strategic sourcing approaches listed in Table 2 and identify the conditions under which they occur, a critical realist case study approach was employed (Easton 2010). This method is powerful in disentangling a set of conditions that generate outcomes and enables a retroductive research process that combines deductive (theory-driven) with inductive (empirically driven) work (Belfrage and Hauf 2017; Easton 2010). In this study, the deductive work involved investigating the existence of the typology of strategic sourcing for ETO manufacturers (Table 2); the inductive work involved creating a deeper understanding of the typology by identifying the conditions associated with its constituent approaches.

In critical realist case studies, analytical generalization is achieved by identifying deep processes at work under contingent conditions (Easton 2010). Therefore, this study employed an embedded, single-case study design (Yin 2018), where the 'case' is an NPD project of a manufacturer operating in the ETO context, while the embedded units are the sourcing processes included in the project. The case study may be described as 'instrumental', as its main focus is to provide insight into phenomena (Stake 2003, 137). Given that in the ETO context, a single NPD project can take many years to complete and include many sourcing processes, this constitutes an adequately rich context for studying the phenomena in question. Furthermore, by comparing different sourcing processes in the same case, there is no need to account for contextual differences among different cases.

4.2. Selection of the case and embedded units

The choice of case context was guided by the theoretical focus and level of access to the company (Eisenhardt and

Graebner 2007; Yin 2018). Therefore, the following selection criteria were used to identify a suitable case context for the study: (1) being an ETO manufacturer, (2) having internal NPD projects, (3) performing strategic sourcing activities (i.e. make-or-buy analysis, supplier selection, supplier collaboration) during NPD and (4) being motivated and willing to provide detailed information. Using these criteria, a large European turbine manufacturer was chosen, in this study named 'TurbineCo' for anonymity reasons.

After gaining access to TurbineCo, the choice of NPD project and embedded units for the study was made through discussions with 'key informants' (John and Reve 1982), which were purposefully selected (Light, Singer, and Willett 1990). These informants included staff from the R&D and purchasing functions because of their knowledge of the NPD projects that could be suitable for investigating sourcing processes. The informants were first informed about the purpose of the study, after which they were asked to identify an NPD project that included many sourcing processes. This resulted in the selection of an NPD project that focused on designing and developing a new turbine for industrial applications. In the project, customer-specific requirements and operating conditions (e.g. pressure, temperature) had to be considered. As an example, application-oriented designs of blades were needed for optimal turbine performance. Meeting such customer requirements required the R&D function to integrate combinations of existing outline designs during NPD, referred to by Gosling, Hewlett, and Naim (2017) as 'adapted design'. Apart from the turbine meeting technical requirements at the right cost, lead time was a major order winner for TurbineCo. This led to the NPD project involving strict promises regarding customer delivery dates. These dates were fixed, making lead time reduction, on-time deliveries, and proactive quality assurance important focus areas.

To manage these challenges, TurbineCo followed an NPD process with three main stages. The first stage was the pre-study stage, which involved identifying and analysing customer requirements, based on which product requirements were specified. Afterwards, the main NPD targets were set, and detailed budgets and plans formulated. The second stage was the concept design stage, which involved designing and evaluating alternative product concepts (regarding the market, technology and financial situation). After choosing the preferred concepts, essential product specifications were documented, reviewed and approved. The third stage was the detail design stage, which involved creating and reviewing detailed documentation required for manufacturing items part of the product. In parallel to the concept and detail stages, the manufacturing and testing of the new product commenced, and experiences from these activities informed product design decisions.

Strategic sourcing played a key role throughout TurbineCo's NPD project. While the manufacturer traditionally single-sourced items or services from small, local suppliers, it now also promoted multiple sourcing from distant suppliers located in low-cost regions. Despite being a large manufacturer, TurbineCo was a relatively small customer of some of these suppliers. Therefore, suppliers were not always equally interested in contributing to NPD, lead time reduction, on-time deliveries and quality. As another challenge, a single turbine consisted of around 5000 items. Changing one of these had many knock-on effects. For example, setting new tolerance requirements for one these items sometimes required new tooling, materials or suppliers. Therefore, the purchasing function of TurbineCo was involved early in the NPD team to manage these sourcing challenges, and included 'purchasing engineers' (Burt and Soukup 1985) with a strong technical background and the ability to work with the R&D function.

After the NPD project selection, workshops were organized at TurbineCo, where the study's purpose and relevant concepts were discussed, and the informants provided further information about the NPD project and had the opportunity to ask questions. Afterwards, the typology (Table 2) was discussed with the informants, and they were asked to identify sourcing processes that reflected the typology's constituent approaches. This resulted in the identification of suitable sourcing processes, of which five were chosen as embedded units in the case study. Table 3 provides information about the embedded units.

4.3. Data collection

For a period of 3 years, empirical data for the case study were collected through interviews and documents. The interviews were conducted with employees who were responsible for or involved in the sourcing processes in the NPD project. The interviewees were selected by using a combination of purposive (Bryman 2016) and snowball sampling (Magnusson and Marecek 2015) techniques. First, purposive sampling involved, together with TurbineCo, identifying individuals that were most likely to yield appropriate and useful information about the focal sourcing processes. Then, snowball sampling involved having the initially selected interviewees recommend potential additional interviewees. The interviewees mainly originated from the R&D and purchasing functions, and more information about the interviewees is shown in Table 4.

The interviews ranged from 30 to 60 min each, involving two to five interviewees at a time, and were audio-recorded and transcribed. The group interview format allowed the interviewees to build on each other's statements by providing viewpoints which may not have been revealed if the interview had been conducted individually. It promoted achieving group-level consensus, while still ensuring that individual interviewees would share their distinct experiences. The interviews were semi-structured and retrospective, encouraging the interviewees to reflect on and make sense of past instances and outcomes of the sourcing processes. Each interview commenced with questions about the interviewees' roles and responsibilities in the NPD project and the five sourcing processes (Table 3). Afterwards, the typology (Table 2) was introduced, and the interviewees were asked to classify the sourcing processes they were involved in according to the typology's five

Table 3. Descriptions of the embedded units.

Embedded unit/ sourcing process	Strategic sourcing approach	Characteristics of the product items	Characteristics of the supplier base
A	EEE	• Items with complex requirements • Items with long lead times • Items requiring extensive supplier input	• Two suppliers • Suppliers with valuable design expertise • Suppliers that were geographically close • Suppliers with high bargaining power • Suppliers that were familiar to TurbineCo
B	ELE	• Items with complex requirements • Items requiring extensive supplier input	• Two suppliers (of which one was selected) • Suppliers with valuable design expertise • Suppliers that were motivated to compete for the same purchase order • Suppliers that were geographically close • Suppliers that were new to TurbineCo
C	EEL	• Items with basic requirements • Items requiring limited supplier input • Items with low volumes	• Two suppliers • Suppliers with low levels of available capacity • Suppliers that were difficult to replace • Suppliers that were geographically close • Suppliers that were familiar to TurbineCo
D	ELL	• Items with basic requirements • Items requiring limited supplier input	• Two suppliers • Suppliers with low bargaining power • Suppliers with high levels of available capacity • Suppliers that were familiar to TurbineCo
E	LLL	• Items with high levels of uncertainty • Items with high volumes	• One supplier • A supplier with high levels of available capacity • A supplier that was geographically distant • A supplier that was familiar to TurbineCo

Table 4. Interviewees.

Interviewee	No. of interviews involved in	Time (h)
Chief Engineer	3	2.5
R&D Manager	2	1.5
Project Manager	2	1.5
Manager Order Execution	2	2
Supply Chain Developer	2	2
Global Demand Planner	3	2.5
S&OP Manager	3	2.5
Strategic Buyer	3	1.5
Head of Production Technology	4	4
Procurement Engineer	5	4.5
Procurement Engineer II	4	3
Procurement Manager	3	2.5
Manager Manufacturing Support	2	2

strategic sourcing approaches (Table 2). Then, the interviewees were asked to reflect on the sourcing processes to identify the conditions that promoted their associated approaches. The interview process concluded when all the sourcing processes were used to populate the typology, and no additional conditions were identified for its constituent approaches. More information about the interviews can be found in Appendix 1.

The retrospective nature of the interviews implies some challenges in relation to memory bias. Thus, to increase the internal validity, respondents were asked to provide documents to enable cross-checks of their statements (i.e. triangulation) (Yin 2018). Examples of such documents include process charts, work procedures and process descriptions of strategic sourcing activities. Furthermore, meetings were organized with the interviewees and their colleagues, which took place online and at TurbineCo. These meetings focused on learning about the interviewees' work context, building trust and testing the concepts used in the research.

4.4. Data analysis

The analysis aimed to investigate the existence of the strategic sourcing approaches listed in Table 2 and the conditions promoting their use. To achieve this, the embedded units (Table 3) were analysed individually, which began by reading the interview transcriptions for each unit several times and writing summary reports (Miles and Huberman 1994). Afterwards, the data analysis followed the analytical coding procedure of King (1998) to extract and refine themes from the interview transcripts and place them into the typology (Table 2).

The first part of the coding process was deductive (theory-driven) and involved organizing data according to the strategic sourcing approaches of the typology. This concerned identifying quotes deemed relevant for confirming the existence of the strategic sourcing approaches and reducing the data to words, sentences and short paragraphs. It should be noted that all the quotes could be used to populate the five approaches of the typology. The second part of the coding process was inductive (empirically driven), which involved using the quotes to interpret the conditions associated with the empirically supported strategic sourcing approaches. This involved identifying higher-level themes by searching for similarities and differences among the quotes. Thus, the themes represent the conditions under which the strategic sourcing approaches occurred, and their identification involved 'retroductive reasoning' (Belfrage and Hauf 2017; Bhaskar 2009) to move from merely a description of the strategic sourcing approaches to explanations of why each of them existed.

Two coders were involved in data analyses to ensure the validity and reliability of the coding. The first coder was responsible for the initial screening for relevant quotes and suggesting themes, while the second coder focused on identifying data gaps to be resolved, verification of the suggested themes, and, if necessary, proposing alternative

themes. Furthermore, the results of the data analysis were shared and discussed with the research participants and, if necessary, modified.

5. Findings

The following sections describe the results of the analysis of the five embedded units in the case study, organized according to the strategic sourcing approaches listed in Table 2 (the complete data structure, including interview quotes, is shown in Appendix 2). The results of the data analysis are summarized in Table 5, with a specific focus on the conditions that promoted the occurrence of the approaches. As seen in the table, each approach has distinct conditions, which are further described in the subsequent subsections.

5.1. EEE

In the NPD project, sourcing process A (Table 3) resembled the EEE approach, and the case study revealed eight conditions that promoted this approach, as shown in Table 5. Two of these conditions promoted an early make-or-buy analysis, three promoted early supplier *selection*, and three promoted early supplier *collaboration*. Concerning the first two conditions, an early make-or-buy analysis occurred because the selection of suppliers and the ramp-up of supplier capacity were expected to require a lot of time. As explained by the Global Demand Planner: 'The make-or-buy analysis for long-lead-time items should already be finished before or during the concept design stage, because it can take years to select suppliers or have their capacity up and running.' In line with the EEE approach, the two suppliers were also selected early during NPD. In TurbineCo, this occurred because there was a need to increase the suppliers' motivation to contribute to NPD. As stated by the Manufacturing Support Manager: 'It is difficult to get commitment from suppliers before buying something. In such situations, we often think that we have dedicated suppliers, but they also work with other customers. We try to avoid this by selecting suppliers early on.' Furthermore, the suppliers were selected early since working with suppliers that may not be selected required excessive resources. The Procurement Engineer explained that this situation could have been avoided if there were more resources dedicated to the comparison of solutions from other suppliers. Finally, the suppliers were selected early because there was little to no competition in the supply market. As argued by the Procurement Manager, late supplier selection is a high-risk endeavour when capable suppliers are scarce due to, for example, complex product requirements.

Also, in line with the EEE approach, the collaboration with the two suppliers started early in the NPD project. This occurred because the suppliers possessed valuable design expertise. As argued by the R&D Manager, this expertise enabled suppliers to help anticipate potential issues and suggest product design changes. Furthermore, early supplier collaboration occurred because the suppliers were responsible for long-lead-time items. As explained by the Procurement Engineer: 'Ties with suppliers that are asked to provide input on long-lead-time items should be established early in NPD. There is otherwise not enough time to properly finish off the design of items and set realistic tolerance requirements.' Finally, early supplier collaboration occurred since there was a need to evaluate the suppliers' ability to contribute to NPD. As stated by the Procurement Manager, early supplier collaboration allowed for determining whether the suppliers had the ability to deliver on their promises.

5.2. ELE

In the NPD project, sourcing process B (Table 3) resembled the ELE approach, and the case study revealed seven conditions that promoted this approach (see Table 5). One of these conditions promoted an early make-or-buy analysis, two promoted late supplier selection, and four promoted early supplier collaboration. Concerning the first condition, an early make-or-buy analysis occurred because there was a need to agree on both internal and supplier capacity requirements. As explained by the Head of Production Technology: 'We divide up responsibilities for workloads and try to manufacture already in concept design, so that everyone agrees on what should be done at home and what should be done outside.'

However, in line with the ELE approach, a single supplier was selected late during the NPD project since its early selection involved too high costs. As argued by the R&D Manager: 'When suppliers are selected early on, they take advantage in terms of price. For example, they raise the price with each small change in an item.' Furthermore, late supplier selection occurred because there was a need to reduce dependence on any one supplier. As stated by the Procurement Manager, early supplier selection could have created lock-in situations in which it is difficult to diversify the supplier base: 'We avoid choosing a supplier early in NPD if it would make it difficult to go to other suppliers later in NPD.'

Nonetheless, in line with the ELE approach, the collaboration with two suppliers started early in the NPD project. This occurred since there was a need to receive input from the suppliers before selecting one of them. As stated by the Procurement Manager: 'We often want to be in contact with suppliers and receive their comments before choosing them.' Early supplier collaboration required the suppliers to be motivated to compete for the same order. The Procurement Engineer argued that without motivation, the suppliers would have been reluctant to provide input and commit resources to a collaboration that may not last. Furthermore, early supplier collaboration occurred because there was a need to evaluate the suppliers' ability to contribute to NPD. The Procurement Engineer explained that since this condition prevailed, candidate suppliers received development contracts, which did not guarantee future purchase orders. The Project Manager argued that early supplier collaboration required the availability of sufficient resources to work with two suppliers: 'If we get a lot of very detailed input from suppliers, but do not place orders with them, we have spent a lot of energy on fine-tuning items to their specific process requirements.'

Table 5. Conditions promoting the strategic sourcing approaches.

Strategic sourcing Approach (embedded unit)	Activity	Promoting conditions	OR/AND
EEE (sourcing process A)	Early make-or-buy analysis	The selection of suppliers requires a lot of time	OR
		The ramp-up of supplier capacity requires a lot of time	AND
	Early supplier selection	There is a need to increase suppliers' motivation to contribute to NPD	OR
		Working with suppliers that may not be selected requires excessive resources	OR
		There is little to no competition in the supply market	AND
	Early supplier collaboration	Suppliers possess valuable design expertise	OR
		Suppliers are responsible for providing input to long-lead-time items	OR
		There is a need to evaluate suppliers' ability to contribute to NPD	
ELE (sourcing process B)	Early make-or-buy analysis	There is a need to agree on both internal and supplier capacity requirements	AND
	Late supplier selection	Early supplier selection involves excessive costs	OR
		There is a need to reduce dependence on any one supplier	AND
	Early supplier collaboration	There is a need to receive input from suppliers before selecting some of them	OR
		Suppliers are motivated to compete for the same order	OR
		There is a need to evaluate suppliers' ability to contribute to NPD	OR
		There are sufficient resources available to work with multiple suppliers	
EEL (sourcing process C)	Early make-or-buy analysis	There is a risk that previously used suppliers are unavailable	OR
		There is a risk of exceeding supplier capacity	AND
	Early supplier selection	There is a need to increase suppliers' motivation to contribute to NPD	AND
	Late supplier collaboration	Most of the design work can be done in-house	OR
		Suppliers have proven their ability to contribute to NPD	OR
		Outsourcing design work would unnecessarily create supplier lock-in situations	
ELL (sourcing process D)	Early make-or-buy analysis	Customer requirements are potentially associated with long-term risks	OR
		There is a need to anticipate and balance internal and supplier capacity	AND
	Late supplier selection	Suppliers are responsible for simple items that are readily available	OR
		There is plenty of competition in the supply market	OR
		There is a need to limit the number of design change requests from suppliers	OR
		There is insufficient information to select suppliers early	AND
	Late supplier collaboration	Suppliers are responsible for basic design work	
LLL (sourcing process E)	Late make-or-buy analysis	There is a need to avoid supplier lock-in situations	OR
		There is insufficient information to perform the make-or-buy analysis early on	OR
		All the required resources are available in-house	AND
	Late supplier selection	There is a risk of exceeding in-house capacity limitations	
	Late supplier collaboration		

5.3. EEL

In the NPD project, sourcing process C (Table 3) resembled the EEL approach, and the case study revealed six conditions that promoted this approach (see Table 5). Two of these conditions promoted an early make-or-buy analysis, one promoted early supplier selection, and three promoted late supplier collaboration. First, as argued by the Global Demand Planner, an early make-or-buy analysis occurred since there was a risk that previously used suppliers were unavailable: 'If we did not establish long-term contracts with suppliers, it may be that some suppliers have ceased to exist.' Furthermore, the make-or-buy analysis was conducted early because there was a risk of exceeding supplier capacity. As argued by the Procurement Engineer, this facilitated the identification and resolution of future supplier capacity constraints.

In line with the EEL approach, two suppliers were selected early during the NPD project. This occurred since there was a need to increase the suppliers' motivation to contribute to NPD. As stated by the S&OP Manager: 'When we develop only a few products in NPD, and suppliers are only responsible for some items of these products late in NPD, we may need to keep them interested by selecting them early.'

However, also in with the EEL approach, the collaboration with the suppliers started late during the NPD project since most of the design work was done in-house. The R&D Manager argued that was possible since most of the needed design expertise was available within the organization. Furthermore, late supplier collaboration occurred since the suppliers had proven their ability to contribute to NPD. As argued by the Procurement Engineer: 'We delay working with suppliers on items when we have good knowledge of their capabilities. This is often the case with local suppliers, especially those with whom we have worked before.' Finally, the collaboration with the suppliers started late since the outsourcing of design work would have unnecessarily created supplier lock-in situations. The Project Manager: 'Supplier collaboration that is too early leads to unnecessarily tailoring a product to supplier needs and limiting the option of selecting other suppliers later in NPD.'

5.4. ELL

In the NPD project, sourcing process D (Table 3) resembled the ELL approach, and the case study revealed seven conditions that promoted this approach (see Table 5). Two of these conditions promoted an early make-or-buy analysis, four promoted late supplier selection, and one promoted late supplier collaboration. First, as argued by the Head of Production Technology, the make-or-buy analysis was conducted early since customer requirements were potentially associated with long-term risks. Furthermore, similar to the EEL approach, the make-or-buy analysis occurred early because there was a need to anticipate and balance internal and supplier capacity. As stated by the Procurement

Engineer: 'Early on, we should try to anticipate what we need later in NPD. For example, in the long term, we may not be able to manage the scope of certain services internally and have to outsource.'

In line with the ELL approach, two suppliers were selected late during the NPD project. As argued by the Procurement Engineer, this occurred because the suppliers were responsible for simple items (e.g. dryers and heaters) that were readily available. The Procurement Engineer also argued that the suppliers were selected late since there was a relatively high level of competition in the supply market. Furthermore, as stated by the Procurement Manager, late supplier selection occurred because there was a need to limit the number of design change requests from suppliers: 'We want to select suppliers as late as possible when we have most of the drawings, specifications, and information. Then we face less of a risk that suppliers will come back and say that they would like to change something or that something was unknown.' Finally, the R&D Manager argued that insufficient design information 'forced' late supplier selection.

In line with the ELL approach, the collaboration with the suppliers also started late during the NPD project since the suppliers were responsible for basic design work. As argued by the Procurement Manager: 'When we are dealing with relatively simple items, like plate bits that have simple processes and metals, we don't really need to worry about the engineering behind the part.'

5.5. LLL

In the NPD project, sourcing process E (Table 3) resembled the LLL approach, and the case study revealed four conditions that promoted this approach (see Table 5). Three of these conditions promoted a late make-or-buy analysis, and one promoted both late supplier selection and collaboration. First, as argued by the Global Demand Planner, the make-or-buy analysis was conducted late to avoid potential supplier lock-in situations: 'It can be preferable to conduct the make-or-buy analysis late. For example, a premature analysis increases the risk of buying critical items or services from suppliers. This leads to becoming dependent on those suppliers for other items or services and limiting the possibility of selecting other suppliers later in NPD.' Furthermore, a late make-or-buy analysis occurred because there was insufficient information to perform the analysis early on. The Global Demand Planner explained that this condition existed since capacity requirements for items were unknown. Finally, as stated by the Procurement Manager, a late analysis was only possible since all the required resources were available in-house: 'We can have a late make-or-buy analysis when we are dealing with items that we can make ourselves but want to keep the door open for suppliers for as long as possible. We really need to be confident that we can make the items ourselves. Otherwise, there is too much risk involved.'

In line with the LLL approach, the selection and collaboration with a single supplier were also postponed until late during the NPD project. As argued by the Procurement Engineer, this occurred since there was a risk of exceeding in-house capacity limitations: 'This happens when we have an item that we can make, and we thought we were going to make, but at the last minute, something happened that made us realise we cannot make it. For example, we unexpectedly run out of capacity. Then we will need to outsource it. This has a knock-on effect on supplier selection and collaboration.'

6. Discussion

As shown in the preceding sections, the five sourcing processes (embedded units) included in the case study support the existence of all the strategic sourcing approaches included in the developed typology (Table 2). Furthermore, the case study reveals that distinct conditions are associated with the approaches. First, the EEE approach may occur when the early NPD stages are highly dependent on scarce suppliers, and substantial effort is required to realise their involvement in NPD. Second, the ELE approach may occur when early supplier input is crucial, but there is also a desire and potential to reduce both supplier dependency and cost. Third, the EEL approach may occur when suppliers are involved late during NPD, but there are only a few capable suppliers available. Fourth, the ELL approach may occur when there is a need to anticipate potential sourcing risks that may occur later during NPD. Finally, the LLL approach may occur when early sourcing is counterproductive, and the benefits of late sourcing outweigh the potential risks.

At a more detailed level, the five approaches each contribute to the NPD sourcing literature. First, the EEE approach (Table 5) supports exiting studies by suggesting that early strategic sourcing allows manufacturers to capitalize on supplier capabilities (e.g. Koufteros, Edwin Cheng, and Lai 2007; Primo and Amundson 2002) or become flexible and have time to prepare and respond to potential supply challenges (Chiu and Kremer 2014). Furthermore, the study also extends the literature by identifying four conditions that promote early strategic sourcing, including (1) a need to increase suppliers' motivation to contribute to NPD, (2) excessive resources needed for collaborating with suppliers that may not be selected, (3) little or no competition in the supply market and (4) a need to evaluate suppliers' ability to contribute to NPD.

The ELE approach (Table 5) supports the study of Wynstra, Van Weele, and Axelsson (1999) by showing that an early make-or-buy analysis saves time and effort in the later NPD stages, as it allows anticipating both internal and supplier capacity requirements. The approach also shows that ETO manufacturers can, if they have sufficient resources to test candidate suppliers and are dependent on their capabilities, collaborate early with suppliers, while delaying the selection some of them until later during NPD. With the ELE approach, suppliers can be selected based on their ongoing contributions and the chances of supplier lock-in situations are reduced. This allows manufacturers to still obtain the benefits that previous research has associated with multiple sourcing (Claypool, Norman, and Needy 2014; Dowlatshahi 1996; Noori and Georgescu 2008).

The EEL approach (Table 5) supports the study of Wynstra, Van Weele, and Axelsson (1999) by showing that an early make-or-buy analysis allows ETO manufacturers to

anticipate and manage supply risks such as limited supplier availability or capacity. The approach further contributes to the literature (e.g. Chiu and Kremer 2014; Noori and Georgescu 2008) by showing that suppliers can be selected early, even though they are expected to contribute with only basic design input to the late NPD stages. This avoids unnecessary supplier lock-in situations while still increasing suppliers' motivation to be part of NPD. It should be noted that the EEL approach may require that suppliers are familiar to the organization and have proven capabilities.

Similar to previous studies (e.g. Wynstra, Van Weele, and Axelsson 1999), the ELL approach (Table 5) shows that an early make-or-buy analysis allows ETO manufacturers to identify whether customer-specific requirements are associated with long-term risks. The approach further adds to previous studies by detailing the conditions that may drive late supplier selection and collaboration. Specifically, while the literature (e.g. Koufteros, Edwin Cheng, and Lai 2007; Primo and Amundson 2002) mainly promotes early supplier input and its benefits, the ELL approach may prevail when design change requests from suppliers should be kept to a minimum or a lack of information obstructs early supplier selection and collaboration. However, this approach may require plenty of competition in the supply market and that suppliers are responsible for basic design input or simple items that are readily available.

Finally, the LLL approach (Table 5) also contributes with new knowledge on strategic sourcing. Specifically, while there are studies focusing on the potential disadvantages of approaching suppliers early (e.g. Hartley, Zirger, and Kamath 1997; Wynstra et al. 2012), this study identifies the conditions that can actually force or warrant late sourcing in the ETO context. This concerns the finding that late strategic sourcing can occur when there is not enough information for an early make-or-buy analysis. In such situations, delaying the make-or-buy analysis until sufficient information becomes available in the late NPD stages reduces the chances of accidentally becoming dependent on suppliers for important items or services. Furthermore, late strategic sourcing can occur when there is a risk of exceeding in-house capacity limitations in the late NPD stages. However, the findings of this study also suggest that late strategic sourcing is may only be recommended when all the resources required for the early NPD stages are available in-house.

7. Conclusions

This study aimed to: (1) conceptualize distinct strategic sourcing approaches that differ in terms of the timing of sourcing activities during an ETO manufacturer's NPD process, (2) investigate their existence and (3) identify their conditions. To achieve these aims, the literature was used to conceptualize five distinct strategic sourcing approaches for ETO manufacturers by making distinctions between early and late timing of the make-or-buy analysis, supplier selection and supplier collaboration. A case study involving a large European turbine manufacturer supported the existence of the five approaches and provided insights into the conditions under which they occur.

7.1. Implications for research

The developed typology of strategic sourcing contributes to the NPD sourcing literature in different ways. First, it contributes by providing a more detailed explanation of the conditions under which ETO manufacturers employ different strategic sourcing approaches (see Table 5). Thereby, the typology provides explanations of when a strategic sourcing approach will occur in the ETO context, and what can cause shifts from one approach to others. This involves insights into how ETO product characteristics influence the choice of a strategic sourcing approach. Specifically, ETO products are associated with uncertainty due to customer-specific requirements that can be difficult to define or may change over time (Gosling and Naim 2009; Hobday, Rush, and Tidd 2000; Willner et al. 2016; Cannas and Gosling 2021). This study suggests that when this uncertainty can be resolved internally and there are plenty of readily available suppliers, early sourcing is counterproductive due to, for example, a risk of supplier dependency. Instead, in such situations, ETO manufacturers are better off postponing strategic sourcing until most design work is done and there is more clarity regarding what input may be needed from suppliers.

As another contribution to the literature, the typology provides clarity on different timing options for strategic sourcing in NPD. Specifically, while previous studies tend to focus either on the early or the late NPD stages, such as the concept design stage (Chiu and Kremer 2014; Noori and Georgescu 2008; Chiu and Okudan 2014) or the detail design stage (Fine, Golany, and Naseraldin 2005; Graves and Willems 2005; Pham and Yenradee 2017; Nepal, Monplaisir, and Famuyiwa 2011), the developed typology provides a more holistic perspective, involving both the early and late NPD stages. The typology also proposes that some strategic sourcing activities (e.g. make-or-buy analysis) can occur early during NPD, while others (e.g. supplier selection or collaboration) may need to be postponed until late during NPD. Thereby, it challenges the implicit assumption prevalent in the literature that early sourcing is always possible or preferred to late sourcing. For example, while the literature suggests that the early integration of suppliers allows access to more or better information earlier in the NPD process by leveraging supplier expertise (Chiu and Kremer 2014; Petersen, Handfield, and Ragatz 2005), this study shows that it also involves a risk of supplier dependency. It further shows that if manufacturers have the resources to perform early design work independently, this risk can be mitigated by postponing supplier selection or collaboration until late in NPD.

7.2. Practical implications

Apart from contributing to the literature, the insights provided by this study also have relevance for practice. Since the purchasing function has become the common interface with suppliers, the literature recommends its early integration into NPD (Reitsma et al. 2023; Mikkelsen and Johnsen 2019; Schiele 2010). This allows the purchasing function to, together with the R&D function, engage in strategic sourcing

and facilitate the involvement of suppliers with desirable capabilities (Shishodia, Verma, and Jain 2022; Li et al. 2023; Reitsma, Hilletofth, Johansson 2023). This study shows that the NPD team of an ETO manufacturer can employ strategic sourcing approaches that are timing-based. The study also shows that a single NPD project in the ETO context can involve the application of multiple strategic sourcing approaches, depending on the conditions that are faced by the NPD team. The NPD team should therefore constantly consider the potential timing of strategic sourcing activities and assess whether there are conditions that warrant their early or late performance in NPD. This can be achieved through regular meetings throughout the NPD process, where the R&D and purchasing functions share their insights and recommendations. For example, this can reveal that late supplier selection or collaboration is preferred because the purchasing function aims to reduce the risk of supplier dependency and knows that several suppliers are readily available, while the R&D function is certain that sufficient internal resources are available during the early NPD stages. Ultimately, such collaboration creates a fit between strategic sourcing, the conditions faced by the NPD team, and the overall NPD goals and objectives.

It should also be noted that the identified strategic sourcing approaches can be formalized and integrated into stage-gate models, which manufacturers often use to manage NPD. For example, the first three stages of stage-gate models (e.g. Ulrich and Eppinger 2016; Cooper and Sommer 2018) typically determine the scope of the new product, its business case and a detailed product design. The gates at the end of each stage filter out work that does not meet predetermined criteria. The three stages and gates can be complemented with the strategic sourcing approaches – as well as their constituent activities. This helps to ensure that strategic sourcing (apart from the product strategy) is also considered by practitioners during NPD.

7.3. Limitations

This study is, however, not without limitations. First, the developed typology has a product-level focus and does not consider the portfolio-dependent nature of strategic sourcing. Specifically, it does not consider the characteristics (e.g. level of engineering complexity) of the different categories of items or services required for the development of a new product. Another limitation is the use of retrospective interviews for data collection in the case study, which could involve memory bias. To reduce such bias, different lines of questioning about the strategic sourcing approaches and data triangulation between interviews were applied.

Other limitations concern the chosen case study design. First, this study used a case study design including one company, which limits the statistical generalizability of the findings. Instead, the study aimed for analytical generalizability by putting the findings into the context of the developed typology (Table 2) and the previous research on strategic sourcing in NPD. Second, the case study was rooted in the critical realist paradigm, where theoretical saturation cannot be fully achieved (Belfrage and Hauf 2017). Therefore, while critical realism allowed explaining the strategic sourcing approaches, the case study findings may be provisional and subject to revision. Third, the case study did not allow for understanding how common the five strategic sourcing approaches are. This was, however, also not the aim of the paper, which rather sought to provide an account of strategic sourcing during an ETO manufacturer's NPD process and show that they occur.

7.4. Future research

Given that this study is the first to investigate the conditions that drive early or late strategic sourcing in the ETO context, there are several avenues for future ETO-oriented research, seven of which are subsequently discussed. First, future research may increase the theoretical saturation of the typology (Table 2) by conducting case studies that further explore the conditions promoting the typology's constituent approaches. Such research could support or extend the findings of this study. Second, future research may investigate the commonness and effects of the five strategic sourcing approaches through survey studies. Third, future research may focus on the types of categories of supply that must be managed during NPD. For example, such research could explore the usefulness of the strategic sourcing approaches discussed in this study for distinct categories of items or services, which can differ in terms of their technical uncertainty (Eisenhardt and Tabrizi 1995), innovativeness (Fisher 1997), impact on the bottom line and associated supply risk (Kraljic 1983), or impact on both the commercial and operational goals of the organization (Cox 2015). Fourth, future research may provide a longitudinal understanding of how strategic sourcing unfolds in the NPD processes of ETO manufacturers. Such research could use real-life observations of meetings where the different timing options for strategic sourcing and their effects on NPD effectiveness or efficiency are assessed. This promises the identification of trade-offs and conflict, which are phenomena that offer major contributions to both the literature and practice (Blackler 1993). Fifth, future research may conduct additional qualitative studies to derive testable hypotheses around the identified strategic sourcing approaches – including their conditions. Hereafter, these hypotheses can be tested through survey studies. Sixth, future research may explore how ETO manufacturers can ensure that appropriate attention is paid to strategic sourcing during NPD – which is a context predominantly involving staff from the R&D function (Reitsma et al. 2023; Willner et al. 2016). Finally, future research may shift from only focusing on ETO manufacturers to also focusing on their first or lower tier suppliers. This may reveal how suppliers perceive and are affected by the strategic sourcing approaches included in the typology of this study.

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Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article.

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Appendix 1. Interview guide

1. Questions regarding the NPD project:
   - What are your roles and responsibilities in the organization?
   - Please describe your expectations of the NPD project.
   - What were the main stages in the NPD project?
   - What were your responsibilities throughout the NPD project?
2. Questions regarding the sourcing processes:
   - In which of the five sourcing processes (Table 3) were you involved?
   - Please provide a general description of the sourcing processes you were involved in.
   - What type of product items or services was involved in the sourcing processes?
   - What type of suppliers was involved in the sourcing processes?
   - What were your roles and responsibilities in the sourcing processes?
   - Please describe how the make-or-buy analysis was conducted in the sourcing processes.
   - Please describe how suppliers were selected in the sourcing processes.
   - Please describe how suppliers were collaborated with in the sourcing processes.
3. Questions that allowed investigating the existence of the strategic sourcing approaches included in the typology (Table 2) and identifying the conditions under which they occurred:
   - Could you describe each sourcing process by following one of the typology's approaches?
   - For each of the sourcing processes, what were the conditions affecting the timing of the make-or-buy analysis in the NPD project?
   - For each of the sourcing processes, what were the conditions affecting the timing of supplier selection in the NPD project?
   - For each of the sourcing processes, what were the conditions affecting the timing of supplier collaboration in the NPD project?
   - Do you have other colleagues who can provide additional insights into the sourcing processes?

Appendix 2. Data structure

	Activity	Condition (the activity occurs when …)	Quote
EEE approach	Early make-or-buy analysis	The selection of suppliers requires a lot of time	'The make-or-buy analysis for long-lead-time items should already be finished before or during the concept design stage, because it can take years to select suppliers or have their capacity up and running'.
		The ramp-up of supplier capacity requires a lot of time	
	Early supplier selection	There is a need to increase suppliers' motivation to contribute to NPD	'It is difficult to get commitment from suppliers before buying something. In such situations, we often think that we have dedicated suppliers, but they also work with other customers. We try to avoid this by selecting suppliers early on'.
			'We sometimes have to select suppliers early on, for them to actually be motivated. Because if you have not selected them yet, they don't really show you everything'.
			'Suppliers usually do not put in that much work until they have the business in hand'.
		Working with suppliers that may not be selected requires excessive resources	'We often don't have the time or resources to work with two suppliers at the same time. It takes a lot of resources to work on different solutions from different suppliers'.
		There is little to no competition in the supply market	'We are forced to select suppliers early on if we are looking at using a process or technique that is new, very specialised, rare, or has very long lead times. For example, when we use certain new coatings or material grades'.
	Early supplier collaboration	Suppliers possess valuable design expertise	'Since 70% of the product cost is put into the concept, it is important to anticipate issues early on and use the suppliers' expertise for more than just delivering a component'.
		Suppliers are responsible for providing input to long-lead-time items	'Ties with suppliers that are asked to provide input on long-lead-time items should be established early in NPD. There is otherwise not enough time to properly finish off the design of items and set realistic tolerance requirements'.
			'We don't want to miss deadlines due to the fact that we are late with receiving supplier input and that suppliers have difficulty implementing what we have decided'.
		There is a need to evaluate suppliers' ability to contribute to NPD	'We also need to start the relationships very early. If we don't do this, we don't know whether they can deliver on their promises'.
ELE approach	Early make-or-buy analysis	There is a need to agree on both internal and supplier capacity requirements	'We divide up responsibilities for workloads and try to manufacture already in concept design, so that everyone agrees on what should be done at home and what should be done outside'"
	Late supplier selection	Early supplier selection involves excessive costs	'When selecting a supplier in concept design, we will not get the most advantageous price when placing orders'.
			'When suppliers are selected early on, they take advantage in terms of price. For example, they raise the price with each small change in an item'.
		There is a need to reduce dependence on any one supplier	'We avoid choosing a supplier early in NPD if it would make it difficult to go to other suppliers later in NPD'.
			'We select suppliers late when we want to spread our chances of success and have two suppliers'.
	Early supplier collaboration	There is a need to receive input from suppliers before selecting some of them	'We often want to be in contact with suppliers and receive their comments before choosing them'.
		Suppliers are motivated to compete for the same order	'It can be preferable to design items internally and send out those designs to two suppliers. This enables us to see who comes back with the best comments before selecting one of them'.
		There is a need to evaluate suppliers' ability to contribute to NPD	'We start the development with two suppliers parallel. And then once we get to a certain level of confidence in one supplier, we drop the other one. We then use development contracts, with does not come with any promises or guarantees that suppliers will be selected'.
		There are sufficient resources available to work with multiple suppliers	'If we get a lot of very detailed input from suppliers, but do not place orders with them, we have spent a lot of energy on fine-tuning items to their specific process requirements'"
EEL approach	Early make-or-buy analysis	There is a risk that previously used suppliers are unavailable	'If we did not establish long-term contracts with previously used suppliers, it may be that some suppliers are no longer available'.
		There is a risk of exceeding supplier capacity	'We clarify from the beginning how we intend to load our suppliers, to avoid overloading them'.

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	Activity	Condition (the activity occurs when …)	Quote
	Early supplier selection	There is a need to increase suppliers' motivation to contribute to NPD	'When we develop only a few products in NPD, and suppliers are only responsible for some items of these products late in NPD, we may need to keep them interested by selecting them early'.
	Late supplier collaboration	Most of the design work can be done in-house	'We are often knowledgeable enough to make sensible product designs without very early supplier intervention. In those cases, suppliers can be involved during the detail design stage'.
			'When we are dealing with build-to-print or very simple commodities, suppliers have limited influence on the design. We wait until we have all the design drawings before we approach the suppliers and ask them to become active'.
		Suppliers have proven their ability to contribute to NPD	'We delay working with suppliers on items when we have good knowledge of their capabilities. This is often the case with local suppliers, especially those with whom we have worked before'.
		Outsourcing design work would unnecessarily create supplier lock-in situations	'Supplier collaboration that is too early leads to unnecessarily tailoring a product to supplier needs and limiting the option of selecting other suppliers later in NPD'.
ELL approach	Early make-or-buy analysis	Customer requirements are potentially associated with long-term risks	'We should start the make-or-buy analysis when assessing customer requirements before the concept design stage. This creates an understanding of what constitutes new and potential long-term risks'.
		There is a need to anticipate and balance internal and supplier capacity	'Early on, we should try to anticipate what we need later in NPD. For example, in the long term, we may not be able to manage the scope of certain services internally and have to outsource'.
			'Sometimes we want to use internal capacity first, before approaching suppliers later in NPD'.
			'Releasing a first tool as early as possible and knowing that you are going to do another one with a supplier later on – the product may not be cheaper in terms of cost but has the potential to come to market much earlier'.
			'Conducting the make-or-buy analysis early on prevents working on items or services that will be bought from suppliers in the long term'.
			'We should avoid internally producing items that will be outsourced in the long run. Otherwise we pay twice for the implementation'.
	Late supplier selection	Suppliers are responsible for simple items that are readily available	'We select suppliers late when they are responsible for small or less important commodities, which we can buy from stock, like heaters or dryers that are readily available at suppliers'.
		There is plenty of competition in the supply market	'We know that there is so much competition in the supply market, that we can easily select suppliers late in NPD'.
		There is a need to limit the number of design change requests from suppliers	'We want to select suppliers as late as possible when we have most of the drawings, specifications, and information. Then we face less of a risk that suppliers will come back and say that they would like to change something or that something was unknown'.
		There is insufficient information to select suppliers early	'We delay supplier selection when we are waiting on the design (to be ready for outsourcing). To really know what it looks like'.
	Late supplier collaboration	Suppliers are responsible for basic design work	'When we are dealing with relatively simple items, like plate bits that have simple processes and metals, we don't really need to worry about the engineering behind the part'.
LLL approach	Late make-or-buy analysis	There is a need to avoid supplier lock-in situations	'It can be preferable to conduct the make-or-buy analysis late. For example, a premature analysis increases the risk of buying critical items or services from suppliers. This leads to becoming dependent on those suppliers for other items or services and limiting the possibility of selecting other suppliers later in NPD'.
		There is insufficient information to perform the make-or-buy analysis early on	'How can we complete the make-or-buy analysis when we do not know how much capacity we need from suppliers'?
		All the required resources are available in-house	'We can have a late make-or-buy analysis when we are dealing with items that we can make ourselves, but want to keep the door open for suppliers for as long as possible. We really need to be confident that we can make the items ourselves. Otherwise, there is too much risk involved'.

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Activity	Condition (the activity occurs when …)	Quote
		'When all the required resources are available, it is so easy for the R&D function – if they have a design – to make it internally the first time and think about suppliers only when really needed later on'.
		'When the R&D function has the knowledge and resources needed for NPD, it is convenient to keep design work internal'.
Late supplier selection Late supplier collaboration	There is a risk of exceeding in-house capacity limitations	'This happens when we have an item that we can make, and thought we were going to make, but at the last minute, something happened that made us realise we cannot make it. For example, we unexpectedly run out of capacity. Then we will need to outsource it. This has a knock-on effect on supplier selection and collaboration'.